Examples of IIR Filter Design

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Key Concepts

1) MATLAB's internal procedure for IIR filter design:

    a) Start with a prototype low pass continuous-time filter.

    b) Pre warp the critical frequencies of the discrete-time filter.

    c) Apply a frequency transformation to convert the prototype filter into the desired pass/stop band configuration.

    d) Apply the bilinear transform to convert the continuous-time filter to a discrete-time system.

2) Using MATLAB to design an IIR filter:

a) Decide on Butterworth, Chebyshev I, Chebyshev II, or elliptic filter design.

b) Two function calls are required.

i) Find the filter order and prototype filter design parameters using buttord, cheb1ord, cheb2ord or ellipord.

ii) Find the filter coefficients using butter, cheby1, cheby2, or ellip.

c) Normalized frequency is used where $[0, 1]$ corresponds to ω on $[0, \pi]$ rads.

d) Passband and stop band ripple are specified in dB using positive values.

3) In general elliptic filters require the lowest order, while Butterworth filters require the highest order to satisfy a given set of specifications.

Video Screenshots

IIR - infinite impulse response

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=1}^{M} b_k x[n-k]$$

Filter Design - find a_k, b_k to satisfy specifications

1) Choose prototype continuous-time lowpass filter $H_{LP}(s)$

2) Prewarp critical frequencies $\Omega = 2\tan\left(\frac{\omega}{2}\right)$

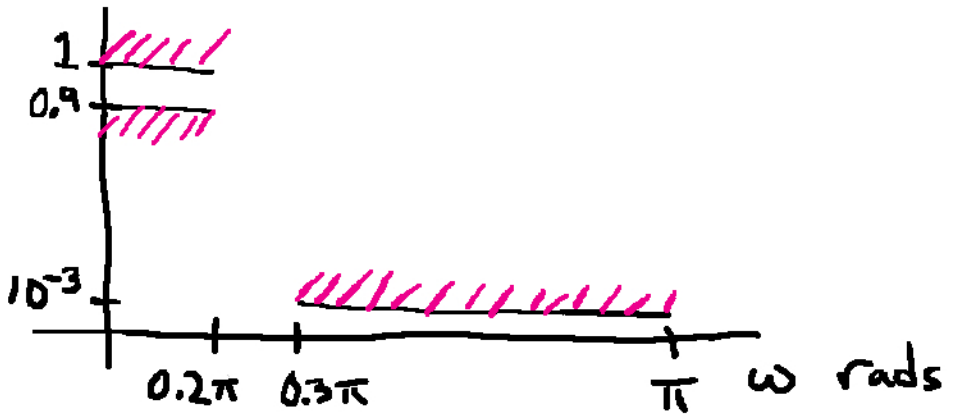
3) Apply frequency transformation

$$H(\tilde{s}) = H_{LP}(s)\Big|_{s=f(\tilde{s})}$$

4) Apply bilinear transformation

$$H(z) = H(\tilde{s})\Big|_{\tilde{s}=2\frac{(1-z^{-1})}{(1+z^{-1})}}$$

Example 1- Low pass



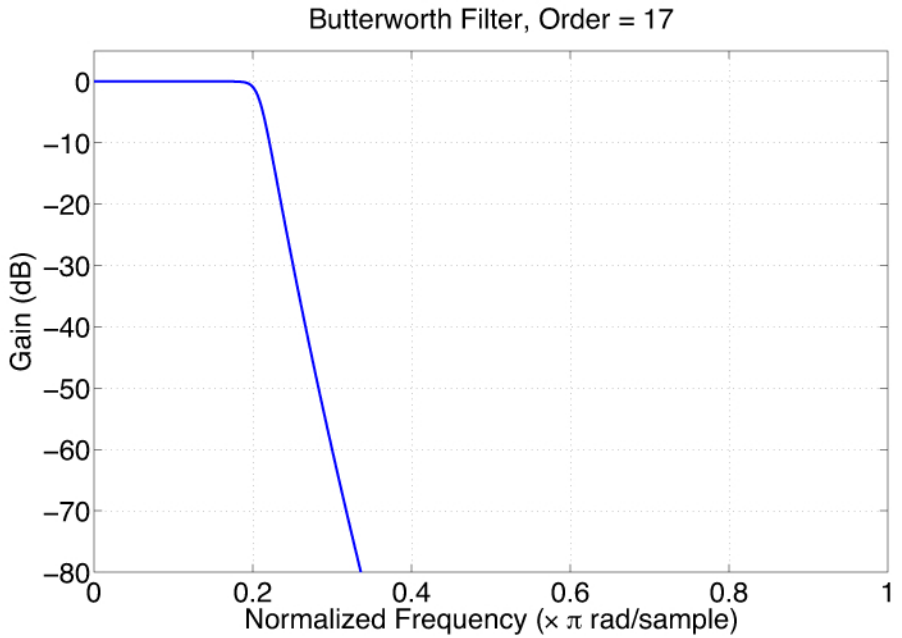
Compare Butterworth, Chebyshev 1 + 2, and elliptic designs

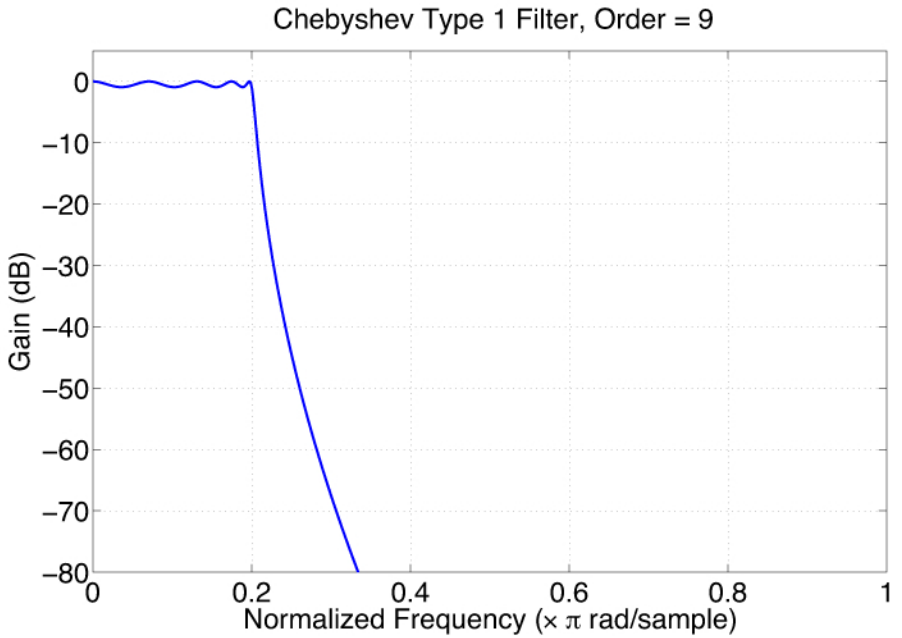
MATLAB: normalized frequency ω/π [0 1]
ripple in dB

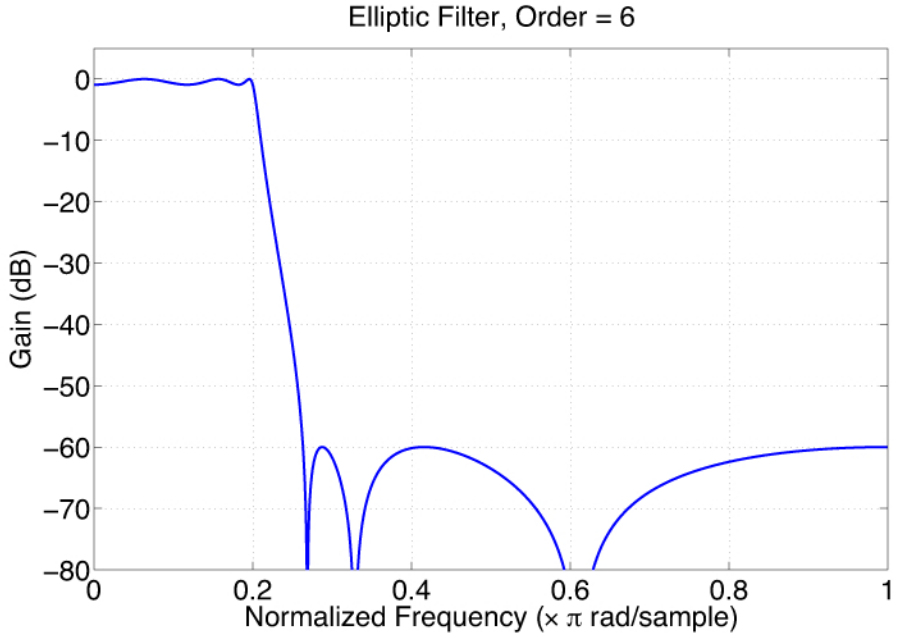
```
passband = 0.2*pi/pi;
stopband = 0.3*pi/pi;
passrip = -20*log10(0.9);
stopatten = -20*log10(0.001);

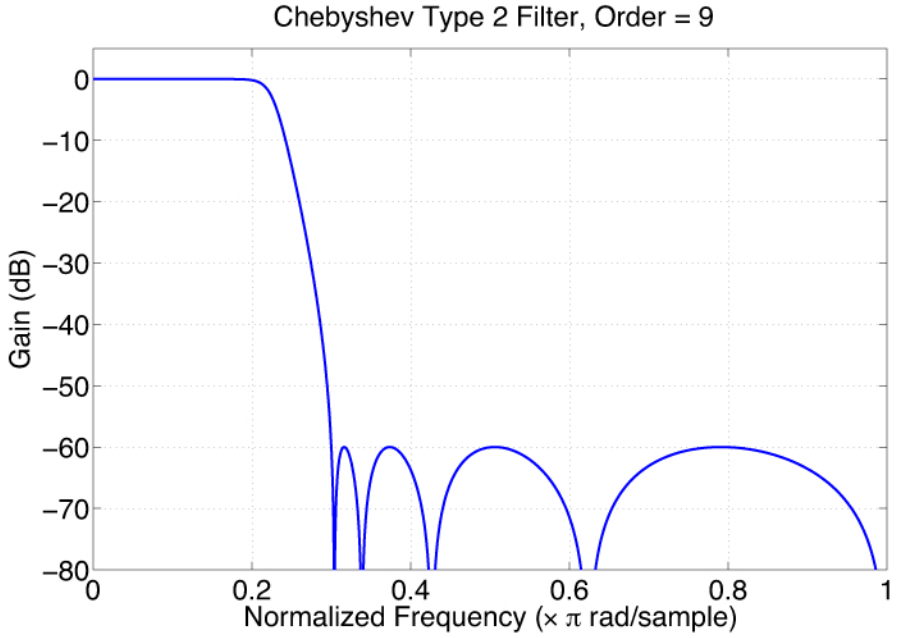
[Nc1, Wnc1] = cheb1ord(passband, stopband, passrip, stopatten);
[Bc1,Ac1] = cheby1(Nc1,passrip,Wnc1);
```

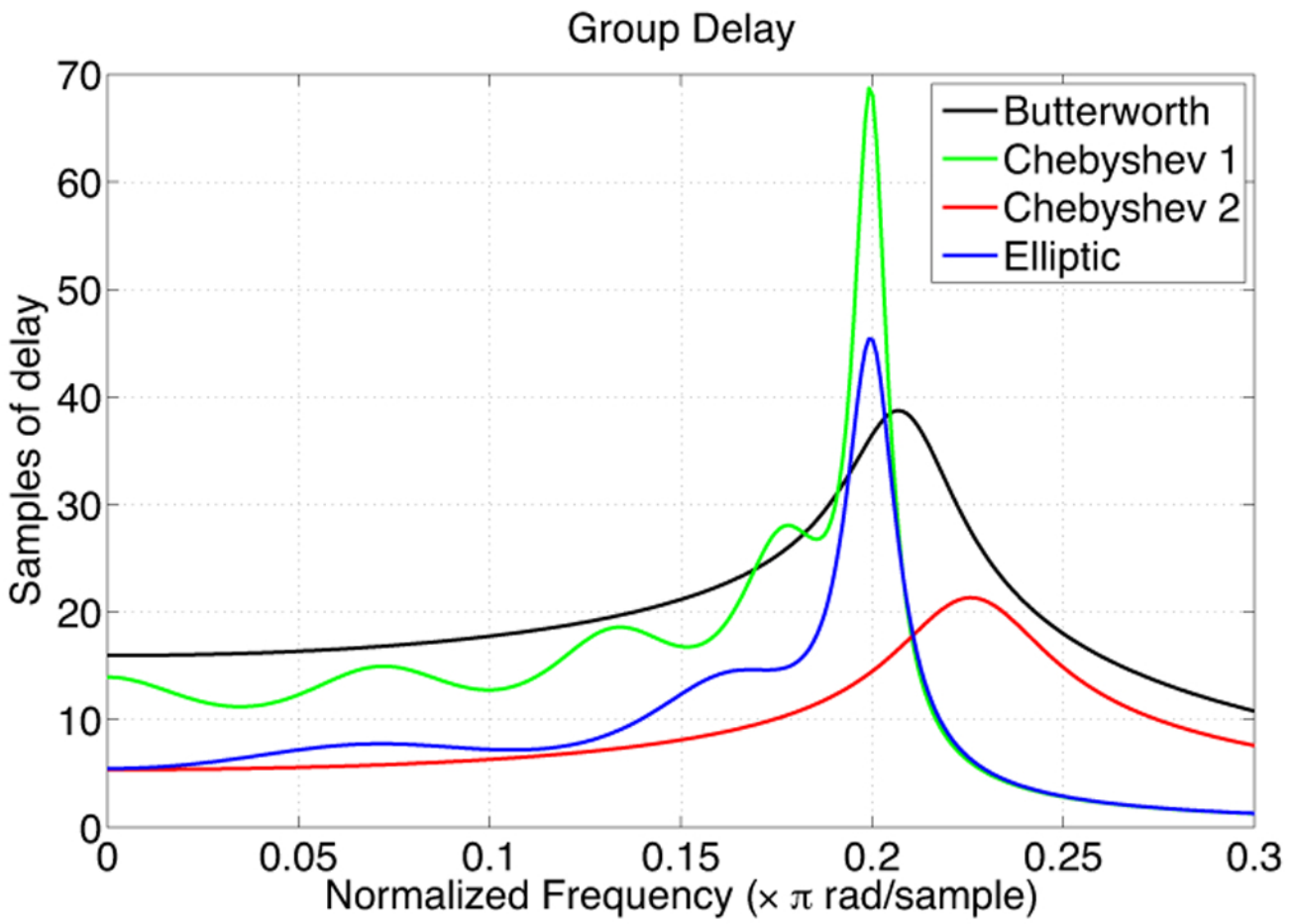
filter order
filter coefficients


Butterworth Filter, Order = 17
Gain (dB)
0
−10
−20
−30
−40
−50
−60
−70
−80
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Chebyshev Type 1 Filter, Order = 9
Gain (dB)
0
−10
−20
−30
−40
−50
−60
−70
−80
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)

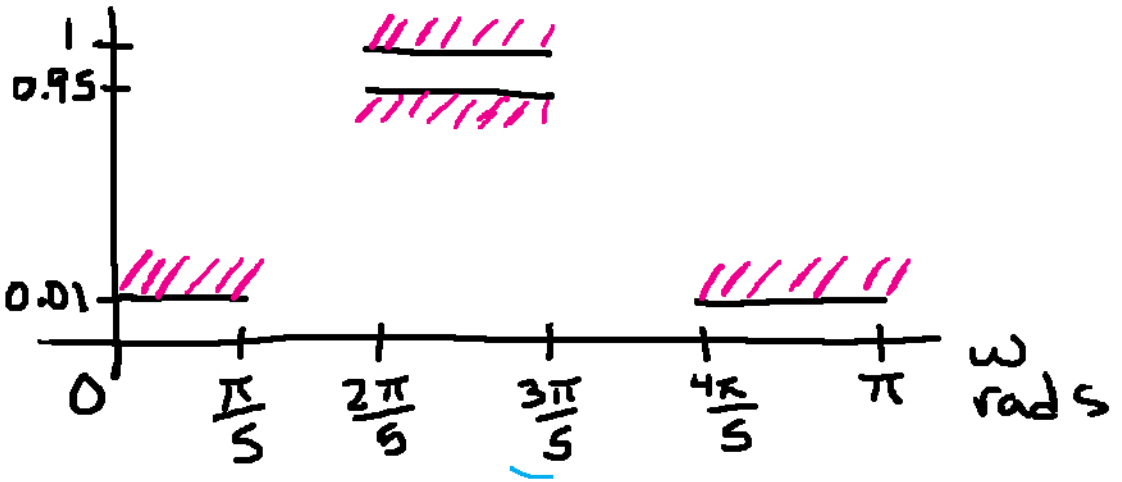

Elliptic Filter, Order = 6
Gain (dB)
0
−10
−20
−30
−40
−50
−60
−70
−80
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Chebyshev Type 2 Filter, Order = 9
Gain (dB)
0
−10
−20
−30
−40
−50
−60
−70
−80
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)



Filter	Order
Butter	17
Cheby1	9
Cheby2	9
Elliptic	6

Example 2: Bandpass



Compare Butterworth and elliptic designs

```
passband = [ 0.4*pi 0.6*pi]/pi;
stopband = [ 0.2*pi 0.8*pi]/pi;
passrip = -20*log10(0.95);
stopatten = -20*log10(0.01);

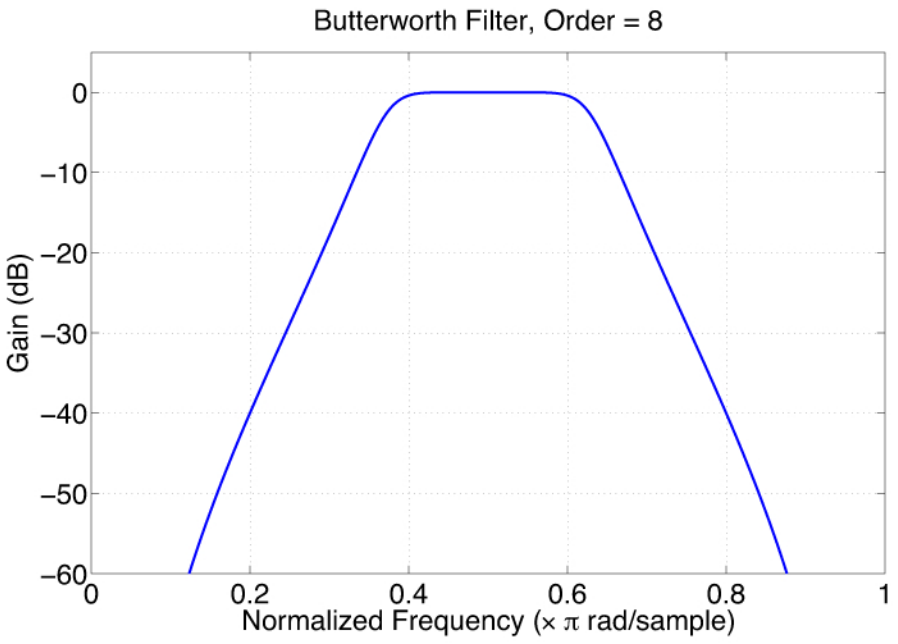
[Ne,Wne] = ellipord(passband,stopband,passrip,stopatten);
[Be,Ae] = ellip(Ne,passrip,stopatten,Wne);
```

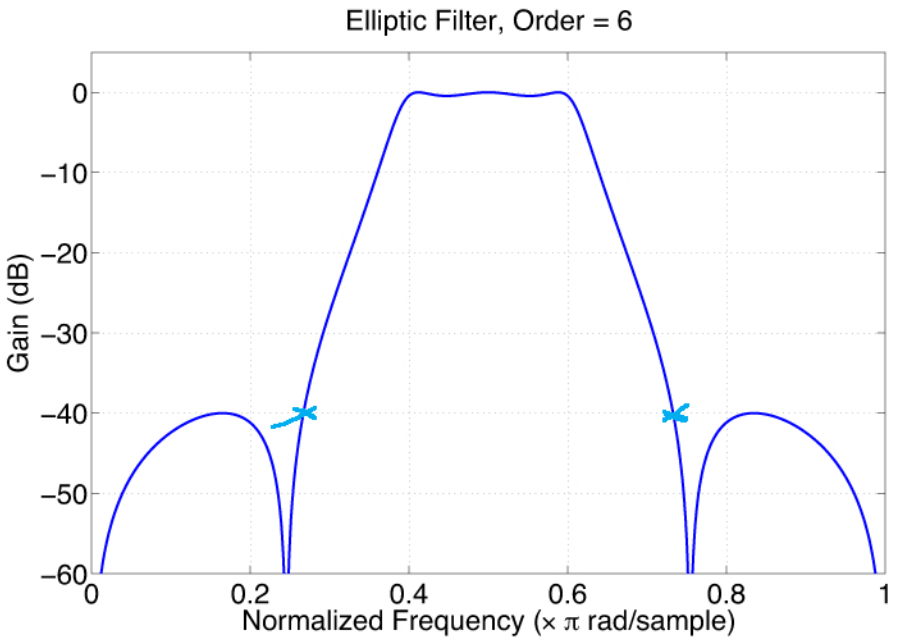
normalize frequency
pass/stopband ripple in dB

filter order
filter coefficients

11:13

6


Butterworth Filter, Order = 8
Gain (dB)
0
−10
−20
−30
−40
−50
−60
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Elliptic Filter, Order = 6
Gain (dB)
0
−10
−20
−30
−40
−50
−60
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)

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